

MultiShopNG

Multi-Vendor E-Commerce & Payment Escrow Platform

OFFICIAL SYSTEM ARCHITECTURE DOCUMENTATION

System Architecture, API Specification & Technical Reference

Comprehensive technical documentation detailing platform architecture, multi-tenant shop customization, 6-digit confirmation code payment escrow protocol, multi-gateway integration (Monnify, Paystack, Stripe), Backblaze B2 cloud storage, and system security standards.

Author & Developer

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Target Environment

Production & Local Dev

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1. Executive Summary & System Overview

MultiShopNG is an enterprise-grade multi-tenant e-commerce platform designed to bridge trust deficits in online shopping through an automated payment escrow system, while empowering independent merchants to create highly customizable digital storefronts.

Key Value Proposition: MultiShopNG combines multi-vendor shop management with a mandatory **6-Digit OTP Delivery Escrow Protocol**. Buyer funds are held securely until the customer verifies package contents and discloses their unique 6-digit delivery code to the merchant.

Core Differentiators

 **Payment Escrow Engine**

100% buyer and seller payment protection against e-commerce scams, non-delivery, and payment disputes.

 **15+ Custom Shop Themes**

Pre-built luxury, modern, and niche storefront templates (HoneySpicy, Obsidian, Artisan, Boho, Cyberpunk, etc.).

 **Multi-Gateway Checkout**

Native integration with Monnify (Moniepoint MFB), Paystack, Stripe, and Direct Bank Transfers.

 **High-Speed Cloud Storage**

Redundant S3 Object Storage hosted via Backblaze B2 for fast global media streaming.

2. System Architecture & Technology Stack

Layer	Technologies & Infrastructure	Purpose & Key Modules
Frontend	React 18, Vite 6, TailwindCSS, Framer Motion, Axios	Single Page Application (SPA), dynamic theme engine, responsive buyer/seller dashboards.
Backend API	Python 3.12, Django 5.x, Django REST Framework (DRF)	RESTful API endpoints, payment webhooks, multi-tenant routing, escrow business logic.
Database	PostgreSQL 16 (Railway Managed)	Relational database storing multi-tenant schema, orders, wallet balances, and audit logs.
Media Storage	Backblaze B2 S3, <code>django-storages[boto3]</code>	High-speed public S3 media storage for product photos, store banners, and logos.

Layer	Technologies & Infrastructure	Purpose & Key Modules
Email Engine	Brevo (Sendinblue) REST API v3	Transactional email delivery for payment receipts, OTP codes, and dispute notifications.
Hosting	Railway (Backend), Netlify/Cloudflare Workers (Frontend)	Automated containerized deployment with zero-downtime SSL edge routing.

3. Core Features & Business Logic

3.1. The 6-Digit Delivery Escrow Protocol

The escrow mechanism prevents e-commerce scamming by keeping buyer payments in a platform-managed holding wallet until verified delivery:

- Checkout & Capture:** The buyer completes payment via Monnify, Paystack, or Stripe. Funds enter `escrow_status = "held"`.
- OTP Generation:** The system automatically generates a unique 6-digit numeric confirmation code (e.g. `849204`) bound to the order group.
- Physical Delivery & Inspection:** The merchant ships the item. The buyer inspects the package upon courier arrival.
- Irreversible Code Release:** The buyer discloses the 6-digit code to the courier/merchant *only* if satisfied.
- Wallet Transfer:** Upon entering the matching code in the seller panel, funds move to the seller's available wallet balance.

3.2. Merchant Tiered Subscriptions & Custom Themes

Plan Tier	Monthly Price	Store & Inventory Limits	Custom Theme Access
Free Starter	₦0 / month	1 Store • 15 Products	Standard Minimalist Theme
Growth Tier	₦5,000 / month	3 Stores • 100 Products	HoneySpicy, Artisan, Boho
Pro Merchant	₦12,000 / month	10 Stores • Unlimited Products	All 15+ Themes (Obsidian, Cyberpunk, etc.)
Enterprise Scale	Custom Quote	Unlimited Stores & Staff Accounts	Custom CNAME Domains & Priority Support

3.3. Multi-Gateway Payment System

MultiShopNG supports seamless payment processing tailored for African and global commerce:

- Monnify (Moniepoint MFB):** Instant Bank Transfers, USSD codes, Dynamic Account Numbers, and Card checkout.
- Paystack:** Card payments, Apple Pay, and local mobile money channels.
- Stripe:** International credit card processing and USD/EUR currency conversions.
- Manual Bank Transfer:** Direct buyer transfers verified via reference code reconciliation.

4. Database Schema & Key Entities

```
User (id, email, password_hash, is_seller, kyc_status)
└─ Shop (id, slug, name, template_id, banner_url, logo_url, rating)
    └─ Product (id, slug, name, base_price, inventory, primary_image)
        └─ OrderGroup (id, order_id, shop_id, status, escrow_status, delivery_code)
            └─ PaymentTransaction (id, reference, gateway, amount, status_code)
```

5. API Endpoint Reference

HTTP Method	Endpoint Path	Description & Access
POST	/api/users/register/	Register a new user account. (Public)
POST	/api/users/login/	Authenticate user and set HttpOnly JWT refresh cookie. (Public)
GET	/api/products/	Fetch product listings with category, price, and shop filters. (Public)
GET	/api/search/	Global search endpoint returning matching products, shops, and facets. (Public)
GET	/api/orders/delivery_codes/	Fetch buyer 6-digit delivery confirmation codes for active orders. (Auth)
POST	/api/orders/confirm_delivery/	Input 6-digit code to release escrow funds to seller wallet. (Seller)
POST	/api/payments/monnify/webhook/	Process instant payment notifications from Monnify gateway. (Public Webhook)

6. Security, Compliance & Data Privacy

- **Authentication Security:** Dual-layer authentication utilizing HttpOnly `SameSite=None; Secure` cookies with Bearer token header fallback for cross-domain requests.
- **Scam Protection & Automated Suspension:** Automatic temporary account suspension triggered if a storefront receives **3+ unique scam reports** within 7 days.
- **KYC Compliance:** Sellers must submit Government IDs, CAC business registration, or BVN verification before requesting wallet payouts.
- **Data Privacy:** Strict adherence to NDPR and GDPR regulations; personal user data is never sold to third-party ad networks.

7. Environment Variables Architecture Template (Sanitized)

```
# Monnify Gateway Configuration (Sanitized Placeholders)
MONNIFY_API_KEY="MK_PROD_YOUR_API_KEY_HERE"
MONNIFY_BASE_URL="https://api.monnify.com"
MONNIFY_CONTRACT_CODE="286935449446"
MONNIFY_SECRET_KEY="YOUR_MONNIFY_SECRET_KEY_HERE"

# Database Configuration
DATABASE_URL="postgresql://postgres:YOUR_PASSWORD@railway.app:5432/railway"

# Transactional Email Engine (Brevo)
BREVO_SENDER_EMAIL="multishopng@apexlabs.it.com"
BREVO_API_KEY="xkeysib-YOUR_BREVO_API_KEY_HERE"

# Backblaze B2 Cloud Object Storage
B2_APPLICATION_KEY_ID="00562ca274a2c5f00000000002"
B2_APPLICATION_KEY="YOUR_B2_APPLICATION_KEY_HERE"
B2_BUCKET_NAME="multishopng"
```

✦ **Maintenance & Security Note:** Production API keys and secret signatures must be stored strictly in environment secret variables (e.g. Railway / Netlify Secrets) and never committed to source control or public documentation.